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THE NATURE OF MJY RADIO SOURCES

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ABSTRACT The properties of milliJansky radio sources are summarized as explored *via* multi-wavelength observations of the 5C12 sample. New optical observations show that several sub-populations are involved, each with attributes which bear on the nature and cosmic evolution of active galaxies. In view of this improved understanding of the fainter sources, the prospects are considered for using the new large-scale sky surveys at 1.4 and 5 GHz for investigations of Universal structure at redshifts far beyond those reached in current large-scale galaxy surveys.

INTRODUCTION

Figure 1 – a source-count at 1.4 GHz – illustrates why the mJy radio sources are relevant to key astrophysical and cosmological questions. There are three major constituents: (A) the brightest sources providing the "Euclidean" portion of the count (Wall and Peacock 1985), nearby sources of moderate to low radio luminosity such as M87 and Cen A; (B) the powerful radio galaxies and radio-loud QSOs whose extreme cosmological evolution has been the subject of numerous detailed investigations, *e.g.* Longair 1966, Wall *et al.* 1980, Condon 1984, Dunlop and Peacock 1990; and (C) the sub-mJy sources, recognized as a new component by Windhorst *et al.* (1985), and identified with starburst galaxies as described in this volume by Benn *et al.*

The mJy sources are at a crossroads. For population B, they can define the luminosity function at lower powers and greater distances; for population C they may include the nearest or most extreme members. As examples of their import: do objects in the transition region suggest that populations B and C are distinct? The question is important for AGN models which derive their nuclear power from starbursts (Terlevich *et al.* 1992). And can the redshift-diminution or cutoff seen by Dunlop and Peacock (1990) be established for lower radio luminosities? What is the cosmic history of FRI objects, a matter of interest

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in unified models in which these lower-power objects are the unbeamed parent population of BL Lac objects (Browne 1983; Padovani and Urry 1990, 1991)?

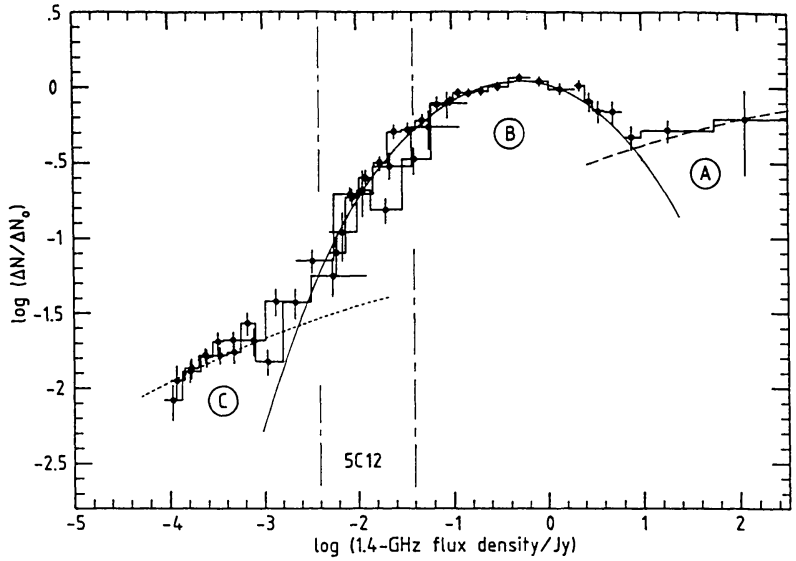

Fig. 1. The source count at 1.4 GHz, in relative differential form (Kellermann and Wall 1987, Windhorst *et al.* 1990). The major three constituents sketched are (A) nearby radio sources of moderate luminosity, (B) radio-powerful QSOs and radio galaxies, and (C) starburst galaxies. The approximate flux-density range occupied by 5C12 is indicated by broken vertical lines.

We first consider the character of the mJy population as revealed by new data on the 5C12 sample. In addition to addressing foregoing issues, the data raise fresh issues, one of which is anisotropy. New large-scale sky surveys at 1.4 and 5 GHz reach to mJy levels; in the second part of the paper we explore whether improved knowledge of the radio-source populations allows direct exploitation of these surveys in searching for large-scale anisotropies and Universal structure.

THE 5C12 SAMPLE

The 5C12 radio/optical survey of a 4°-diameter field near the NGP reaches S_{408} = 10 mJy and m_v = 22^m.5; data for 300 sources include radio maps (Benn *et al.* 1982, 1988a), radio spectra (Benn *et al.* 1984), photographic photometry (Grueff *et al.* 1984) and IR photometry (Benn *et al.* 1988b). The original identification programme (Benn *et al.* 1988a) yielded a sample of 73 complete and reliable identifications: 30 red galaxies (presumed ellipticals), 7 blue galaxies, 33 stellar (mostly blue) objects and 3 bright (m_v < 15) galaxies. We have now obtained low-dispersion spectra of many of these with the Faint-Object Spectrograph of the William Herschel Telescope. In addition we have obtained deep CCD images for a complete sample of previously-unidentified 5C12 sources to search for optical counterparts below the limit of the photographic survey.

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Optically-passive sources

Optical observations of the 11 brightest red-galaxy identifications were described by Rixon *et al.* (1991). The redshifts are $0.12 < z < 0.31$, placing radio luminosities in the range $24.2 < \log P_{408}(\mathrm{W/Hz}) < 26.3$ ($H_0 = 50\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$); the radio maps suggest that the majority are FRII in character. The galaxies are remarkably homogeneous in showing no evidence of nuclear activity or recent star formation, no significant change of rest-frame colour with redshift, and only small scatter in absolute magnitude, $0^m.4$ about a median $M_V = -22^m.8$, which does not differ significantly from that of the 3CR FRII sources whose radio powers are some two orders of magnitude greater.

Optically-active sources

Two-thirds of the identifications with $m_v < 22^m.5$ have optical characteristics distinguishing them from the passive ellipticals. There is considerable variety, with the different kinds of object loosely separated by radio power. At $\log P_{408}(\mathrm{W/Hz}) > 27$ there are a few 'classical' QSOs, with spectra dominated by non-thermal continua. At $26 < \log P_{408} < 27$, the supposed QSOs have radio-galaxy spectra: all have $M_V \sim -22^m.8$, most show narrow emission lines and a few show broad lines. At yet lower radio powers, there are Seyfert-type objects as well as two blue galaxies which appear to be starbursting systems. Two of the three brightest galaxies are interacting systems, and one of these two is an IRAS source.

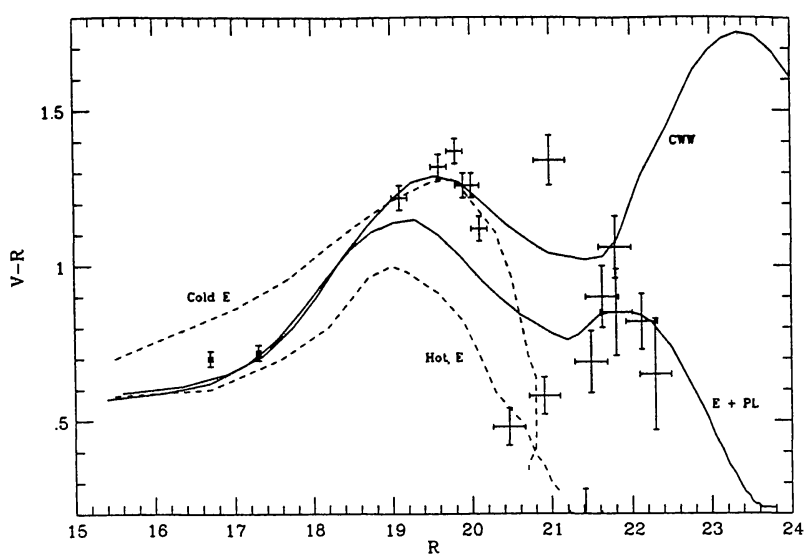

Fig. 2. Colour vs magnitude for the deep identifications from 5C12. Solid curves represent the loci of redshifted ($0.05 < z < 1.5$) elliptical galaxies (a) with spectrum from Coleman *et al.* (1980), and (b) with this spectrum to which a weak power-law continuum has been added (5% at 5500Å with $\alpha = 0.0$). Dashed lines represent the loci of models from Guiderdoni and Rocca-Volmerange (1987). The points at magnitudes brighter than $R = 20^m$ are the passive red galaxies discussed earlier.

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Deep identifications

Thirty new identifications were found in I-band CCD exposures at the positions of a complete sample of 64 previously-unidentified 5C12 sources with $10 < S_{408} < 50$ mJy. V and R magnitudes were obtained for a subset of these objects; these magnitudes suggest that most have redshifts between 0.7 and 1.0 (and hence $\log P_{408} < 26$) if they are no brighter than giant elliptical galaxies. One and perhaps two of the objects are variable, suggesting that they are AGNs. The faint identifications are bluer than local early-type galaxies, due either to young stars or some non-thermal continuum. The magnitudes and colours are most successfully reproduced by a model of an AGN which adds a weak power-law component to a non-evolving elliptical galaxy (Fig 2). Calculation of space density for these objects based on the estimated redshifts is consistent with the hypothesis of *no evolution* ($\log P_{408} \lesssim 26$) from $z = 0$ to the shell $0.7 < z < 1.0$.

Stratification of source-type with radio power

The transition of the characteristic source type from optically-passive galaxy through weak-cored AGN to classical QSO occurs in the same sequence, and at roughly the same powers, in bright-source surveys (*e.g.* 3CR – Laing *et al.* 1983; 2.7-GHz all-sky – Wall and Peacock 1985) and in 5C12 identifications with $m_v < 22$, despite the large difference median redshifts. In particular the lower limit in radio power of radio-loud QSOs (Kellermann *et al.* 1989, Dunlop *et al.* 1989) appears constant with redshift out to $z \sim 0.7$. The deep 5C12 identifications are consistent with the continuation of a redshift-independent 'weak-AGN' stratum out to at least $z = 1$. Thus source type appears to be a function of radio power rather than redshift. This stratification in the $P - z$ plane is important for models of luminosity evolution for radio sources: unless the optical appearance of sources is contrived to change in step with the radio power, such models cannot reproduce it.

SKY DISTRIBUTION AND UNIVERSAL STRUCTURE

Although formal statistical analyses of sky distributions in major radio surveys (*e.g.* Webster 1977) failed to reveal significant deviation from a random and independent uniformity, Shaver and Pierre (1989) showed that the brightest sources with $z < 0.02$ (population A, Fig 1) are strongly clustered to the supergalactic plane. The result explained lingering apparent anisotropies on the very largest scales, $\theta > 10°$ (Pauliny-Toth 1977, Wall 1977). The sub-mJy sources (population C, Fig 1) have smaller median redshift than the mJy and brighter sources (Benn *et al*, this volume), and anisotropy in their distribution on arcmin scales was suggested by Windhorst *et al.* (1990). Anisotropies amongst the powerful radio galaxies and radio-loud QSOs (population B, Fig 1) would be of particular interest in tracing Universal structure on scales out beyond $z \approx 0.2$, the limit of large-scale galaxy surveys. Indeed suggestive deviations from uniformity are present in 5C12. A possible supercluster at $z \approx 0.8$, mapped by the deep identifications, has been discussed by Wall (1990); a second example is the sky distribution of the bright galaxy identifications (Fig 3).

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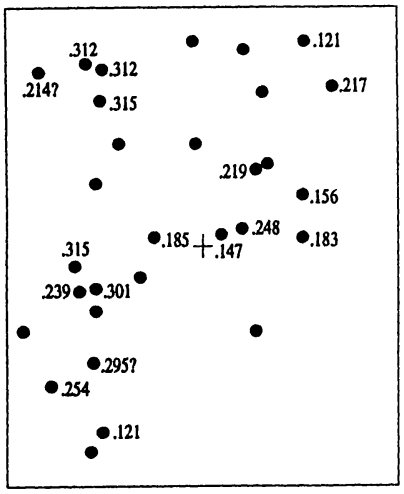

Fig 3. Positions, redshifts (when available) for the 30 galaxy identifications in 5C12 with $m_v \leq 22^m.5$; the field is 4° (α) by 5° (δ) and the cross marks $\alpha = 12^h58^m43^s$, $\delta = 35°14'$. Note the 'redshift systems' suggesting coherent structures across the field.

The new surveys

Prospects for detecting anisotropy have improved with new radio surveys at 1.4 GHz (White and Becker 1992, and references therein) and 4.85 GHz (Becker *et al.* 1991, Gregory and Condon 1991) covering the northern sky down to 25 mJy at 5 GHz and 100 mJy at 1.4 GHz (Fig 4). The surveys come within a factor of 10 of the surface density and equivalent flux density of the 5C12 survey, and they are almost identical in these respects to the selected-area surveys of Wall *et al.* (1971) whose constituents have been so thoroughly studied by Dunlop *et al.* (1989). There is thus a substantial database on which to base interpretation of the catalogue statistics.

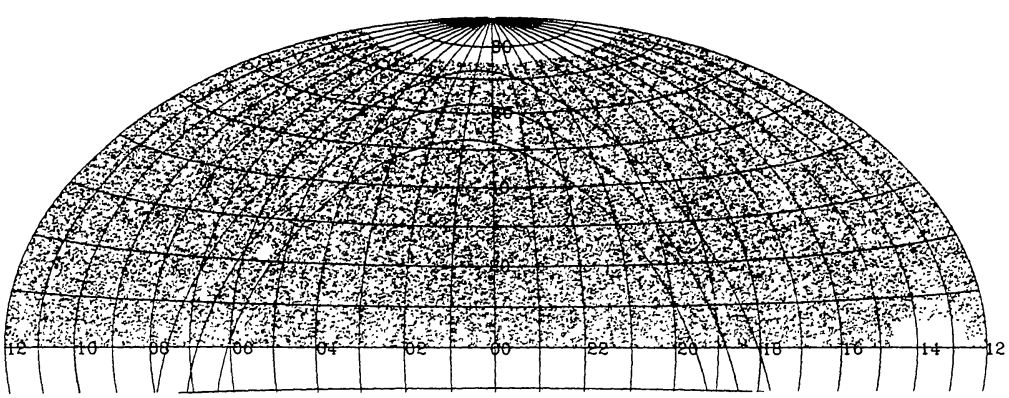

Fig. 4. The northern sky: 54579 sources from the 4.85-GHz catalogue of Gregory and Condon (1991).

Luminosity indicators; selection effects; sky distribution

Could simple 'luminosity indicators' from radio data be invoked to "place" the sources in redshift shells narrow enough to detect structure, without the labour of identification and redshift measurement? There are two such indicators from integrated radio spectra alone, both of very low resolution: — the $P - \alpha$ effect (Heeschen 1960, Laing and Peacock 1980), which for extended radio sources is a correlation between radio luminosity and steepness of spectrum; and the relation

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between spectral shape and identification which distinguishes QSOs from galaxies by virtue of the dominance of high-surface-brightness components in QSOs (*e.g.* Wall 1975).

White and Becker (1992) found that some 20000 sources in the 1.4-GHz catalogue had counterparts in the Texas 365-MHz catalogue *and* the Becker *et al.* (1991) 4.85-GHz catalogue. Fig 5a, a radio colour-colour diagram, shows the form of the radio spectra of these sources, using spectral indices calculated from flux densities at the three frequencies. The equivalent diagram for the 233 sources of the 2.7-GHz all-sky catalogue (Wall and Peacock 1985) is shown in Fig 5b. Optical counterparts (and redshifts) of almost all these sources are known, and in some areas of the diagram ($\alpha_{HF} > 0.0$ for example), there is almost complete mutual exclusion between QSOs and radio galaxies. The $P-\alpha$ effect concerns basically the area in the square $-0.6 > \alpha > -0.9$. The effect is just discernible in two-colour plots of the Wall-Peacock sample for different radio luminosities, but it is below the resolution limit of the spectral data in the 1.4-GHz catalogue. A striking difference between the diagrams is the large numbers of sources below the power-law line in Fig 5a. This may represent a larger proportion of sources with spectral peaks towards the lower frequencies; but perhaps a more likely explanation is underestimation of 365-MHz flux densities due to the interferometric nature of the Texas survey.

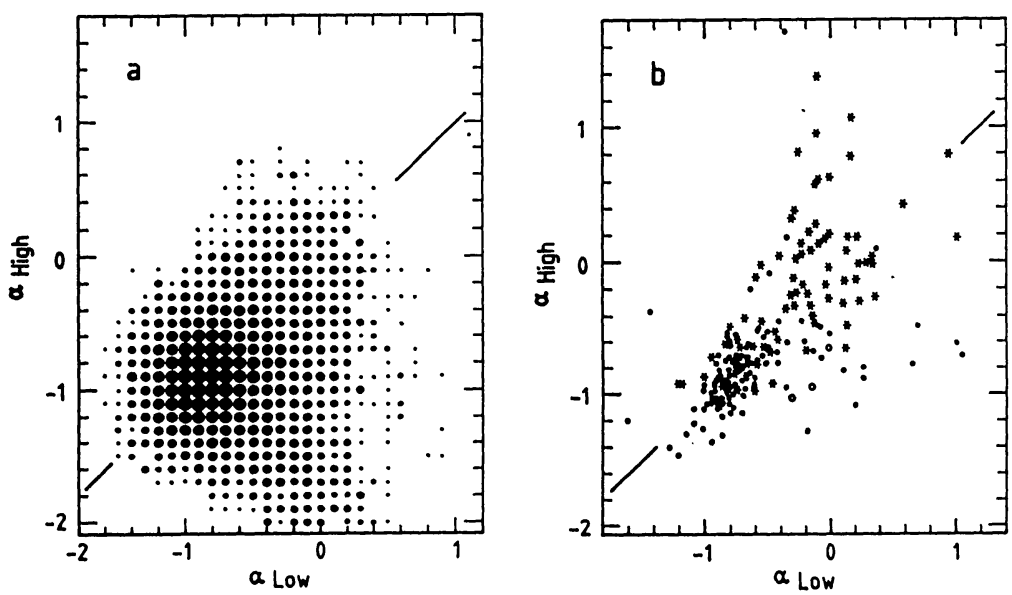

Fig. 5. Radio two-colour diagrams for (a) the White – Becker (1992) catalogue at 1.4 GHz, and (b) the Wall – Peacock (1985) catalogue at 2.7 GHz, with symbols • – galaxy, $*$ – QSO. In (a), the three frequencies at which flux densities were measured are 0.365, 1.4 and 4.85 GHz; in (b), 0.408, 2.7 and 5.0 GHz. The spectral indices are in the sense $S \propto \nu^{\alpha}$.

The comparison thus suggests that (a) selection effects are present in the catalogue data (to which attention has been drawn by the compilers); (b) the $P-\alpha$ effect can provide little luminosity/redshift selection with the present data; but (c) moderately pure samples of QSOs and radio galaxies can be selected on the basis of radio spectral shape alone.

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From these considerations, some subsets of the 1.4-GHz catalogue were constructed to carry out anisotropy searches *via* computation of the two-point correlation function $w(\theta) = N(\theta)/N_r(\theta) - 1$. In all subsets, the region within 10° of the Galactic plane was excised, as were areas of obvious incompleteness in sky plots of the catalogues. A first observation is that small-amplitude signal $w(\theta) \approx 0.05$ is present on very large scales, $\theta > 10°$. This may be due to including some regions of unrecognized incompleteness, or to inconsistencies in scaling and/or flux-density limits over large areas. Of several subsets examined at different flux-density levels and selected as 'QSOs', 'powerful (or weak) radio galaxies' from the radio spectral criteria outlined above, only two showed possible signal, both on scales of $\leq 3°$: a sample corresponding to lower-luminosity radio galaxies (power-law spectra towards the low-luminosity side of the $P-\alpha$ relation), and the sample of sources indicated as extended in the catalogue (Fig 6). It is encouraging that the only suggestions of signal concern samples anticipated to be the nearest; but it is essential to establish a full understanding of catalogue parameters before continuing this exploration.

The clearest path to exploring structure at $z < 0.5$ is identification of the optically-passive ellipticals like those found in 5C12, for which mass photographic photometry can yield appropriate redshift estimates.

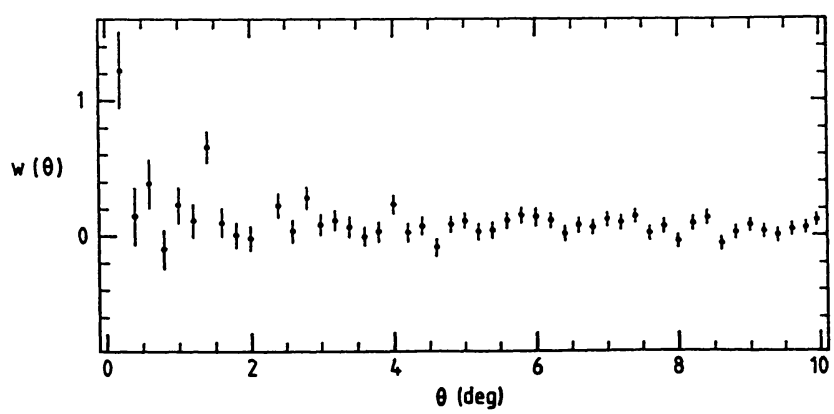

Fig. 6. The two-point correlation function for sources in the 1.4-GHz White – Becker catalogue (excluding the Galactic plane) indicated as extended to the 12′ beam.

SUMMARY

Optical investigations of the 5C12 identifications indicate the presence of (a) optically-passive red galaxies of predominantly FRII radio morphology, whose optical homogeneity makes them useful 'standard candles', and (b) optically-active galaxies with all types of radio structures and with activity ranging from weak starbursters through Seyferts to luminous QSOs. The objects demonstrate a stratification in the $P - z$ plane in which radio power 'selects' host-type independently of redshift. The surface distribution suggests anisotropy; non-uniformities are also present in the redshift distribution. Sky distributions in the new large-area deep surveys also suggest some anisotropies amongst nearer sources, but selection effects are present which require exploration first.

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DISCUSSION:

Hill : Simon Lilly and I have spectra of a dozen fainter red 5C12 sources with $0.35 < z < 0.65$ and half show $[OII]3727$ emission, so the tendency for the lower redshift red id's to have no emission lines does not continue to higher redshifts. It would be interesting to compare those spectra with yours.

Wall: I agree. An analysis of our joint data-sets should indicate whether there is a continuum in the radio–power – optical emission–line strength plane, or whether there is a threshold.

Sandage: What are the absolute magnitudes of the E galaxies you have identified with your 5C mJy sources? Are they fainter than first ranked cluster E's? Are they similar to the E galaxies associated with Jy 3C sources? If similar, here we have 10^2 range in radio power and only about a factor 2 in optical. If so, then it still takes a massive E galaxy parent mass to produce a radio source?

Wall: Yes; the mean is about 0.5 mag less luminous than the 3C E galaxies, a remarkably small change given the change in radio luminosity, as you point out.

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